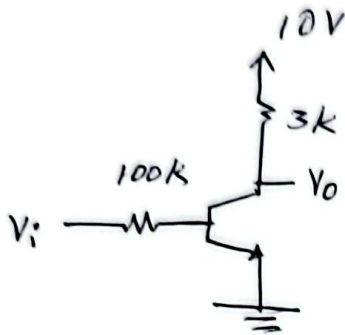


BJT MAS Math

①



$$\beta = 100$$

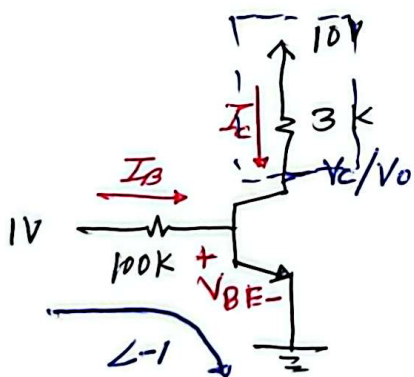
$$V_{BE} = 0.7 \text{ [Active \& Sat.]}$$

$$V_{CEsat} = 0.2$$

② $V_i = 1V$ ③ $V_i = 5V$.

Note, Since for both 1 & 5V is higher than 0.7
let's not assume cut off as $V_{BE} > 0.7$ $[V_{BE} = V_B - V_E^0 = V_B]$

② Assume active,



so in active mode,

$$I_C = \beta I_B$$

$$V_{BE} = 0.7$$

KVL Along L-1,

$$-1 + 100k I_B + V_{BE} = 0$$

$$\Rightarrow 100k I_B = 0.3$$

$$\Rightarrow I_B = 3 \mu A$$

$$I_C = \beta I_B = 300 \mu A = 0.3 mA$$

BJT is in Active Mode.

Ohm's Law,

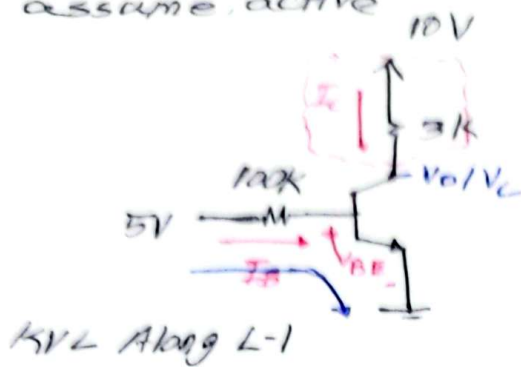
$$\frac{10 - V_C}{3k} = I_C$$

$$\begin{aligned} \Rightarrow V_C &= 10 - 3k I_C \\ &= 10 - 3 \times 0.3 \\ &= 9.1 \end{aligned}$$

$$\begin{aligned} V_{CE} &= V_C - V_E \\ &= 9.1 - 0 \end{aligned}$$

$$V_{CE} = 9.1 > 0.2$$

⑥ assume active



KVL Along L-1

$$-5 + 100k I_B + \frac{V_{BE}}{0.7} = 0$$

$$\Rightarrow 100k I_B = 4.3$$

$$\Rightarrow I_B = 43\mu A$$

$$I_C = \beta I_B = 4300\mu A \\ = 4.3mA$$

Ohm's law,

$$\frac{10 - V_C}{3k} = I_C$$

$$\Rightarrow 10 - V_C = 4.3 \times 3$$

$$\Rightarrow V_C = 10 - 12.9$$

$$\therefore V_C = -2.9$$

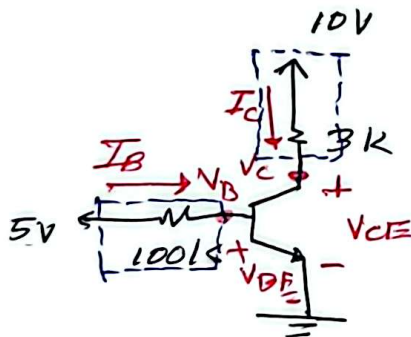
$$V_{CE} = V_C - V_E$$

$$= -2.9 - 0$$

$$V_{CE} < -2.9$$

Not in active mode.

assume, saturation.



$$V_{CE} = 0.2$$

$$\Rightarrow V_C - V_E = 0.2$$

$$\Rightarrow V_C - 0 = 0.2$$

$$\therefore V_C = 0.2$$

$$V_{BE} = 0.7$$

$$\Rightarrow V_B - V_E = 0.7$$

$$\Rightarrow V_B - 0 = 0.7$$

$$\therefore V_B = 0.7$$

$$V_{CE} = 0.2$$

$$V_{BE} = 0.7$$

Ohm's Law,

$$\frac{10 - V_C}{3k} = I_C$$

$$\Rightarrow 10 - V_C = I_C 3k$$

$$\Rightarrow 10 - 0.2 = I_C 3k$$

$$\Rightarrow I_C = \frac{9.8}{3k}$$

$$\therefore I_C = 3.26667$$

$$\frac{5 - V_B}{100k} = I_B$$

$$\Rightarrow 5 - V_B = I_B 100k$$

$$\Rightarrow 5 - 0.7 = I_B 100k$$

$$\Rightarrow I_B = 4.3/100k$$

$$I_C = 3.2667$$

$$I_B = 0.043$$

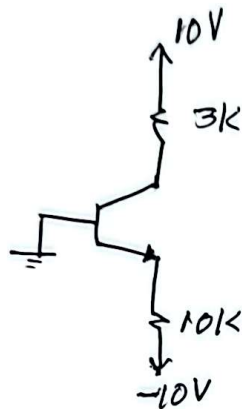
$$\beta_{sat} = \frac{I_C}{I_B} = \frac{3.2667}{0.043} \approx 76$$

$$\beta_{sat} < \beta_{Active}$$

$$\therefore 76 < 100$$

so it is in saturation.

③



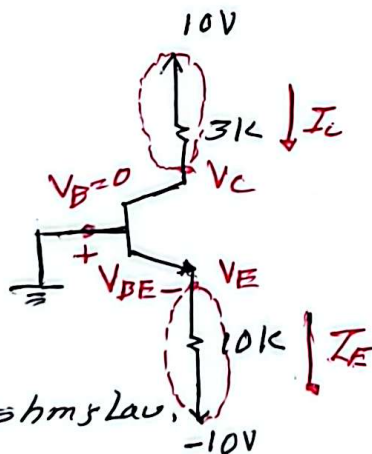
$$\beta = 100$$

$$V_{BE} \approx 0.7$$

$$V_{CE} = 0.2_{sat}$$

Ans

assume active,



now, Ohm's law,

$$\frac{V_E - (-10)}{10k} = I_E$$

$$\frac{-0.7 + 10}{10k} = I_E$$

$$V_{BE} = 0.7$$

$$\Rightarrow V_B - V_E = 0.7$$

$$\Rightarrow 0 - V_E = 0.7$$

$$\Rightarrow V_E = -0.7$$

$$I_E = \frac{9.3}{10k} = 0.93 \text{ mA}$$

during Active

$$I_E = I_C + I_B$$

$$\Rightarrow I_E = \beta I_B + I_B$$

$$\therefore I_E = (\beta + 1) I_B$$

$$I_B = \frac{I_E}{\beta + 1}$$

$$\Rightarrow I_B = \frac{0.93}{100+1}$$

$$\therefore I_B = 9.207 \mu A$$

$$I_C = \beta I_B = 100 \times 9.207 \\ = 0.92 \text{ mA}$$

$$\frac{10 - V_C}{3K} = I_C$$

$$\Rightarrow 10 - V_C = 0.92 \times 3$$

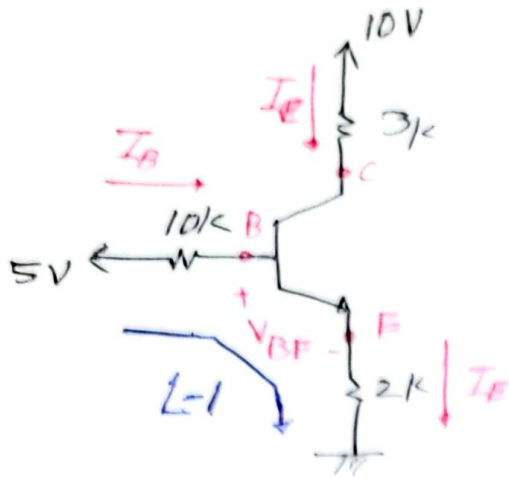
$$\therefore V_C = 7.24$$

$$V_{CE} = V_C - V_E \\ = 7.24 - (-0.7) \\ = 7.94$$

$$V_{CE} = 7.94 > 0.2$$

$\therefore$ BJT is in active mode.

3



$\beta = 100$
 $V_{BE} = 0.7$
 $V_{CE(sat)} = 0.2$

assume active, $I_E = (\beta + 1) I_B$

KVL Along L-1,

$$-5 + 10k I_B + \underbrace{V_{BE}}_{0.7} + \boxed{2k I_E} = 0$$

$$\Rightarrow 10k I_B + 2k (\beta + 1) I_B = 4.3$$

$$\Rightarrow 10k I_B + 2k \times 101 I_B = 4.3$$

$$\Rightarrow 212k I_B = 4.3$$

$$\therefore I_B = 20.28 \mu A$$

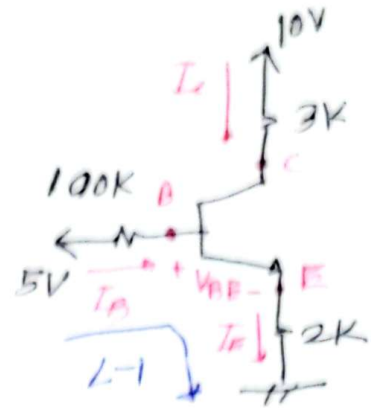
now, $I_C = \frac{10 - V_C}{3k}$

$$\Rightarrow \beta I_B = \frac{10 - V_C}{3k}$$

$$\Rightarrow V_C = 10 - \beta I_B 3k$$

$$\Rightarrow V_C = 10 - 100 \times 20.28 \mu A \times 3k$$

$$\therefore V_C = 3.916$$



assume active,

KVL along L-1,

$$-5 + 100k I_B + V_{BE} + 2k I_E = 0$$

$$\Rightarrow 100k I_B + 2k (\beta + 1) I_B = 4.3$$

$$\Rightarrow 100k I_B + 2k \times 101 I_B = 4.3$$

$$\Rightarrow 302k I_B = 4.3$$

$$\therefore I_B = 14.23 \mu A$$

now,

$$I_C = \frac{10 - V_C}{3k}$$

$$\Rightarrow \beta I_B = \frac{10 - V_C}{3k}$$

$$\Rightarrow V_C = 10 - \beta I_B 3k$$

$$\Rightarrow V_C = 10 - 100 \times 14.23 \times 10^{-6} \times 3000$$

$$\therefore V_C = 5.731$$

$$\therefore I_E = \frac{V_E - 0}{2K}$$

$$\Rightarrow V_E = I_E 2K$$

$$\Rightarrow V_E = (\beta + 1) I_B 2000$$

$$\Rightarrow V_E = (100 + 1) I_B 2000$$

$$\Rightarrow V_E = 101 \times 2000 \times 20.28 \times 10^{-6}$$

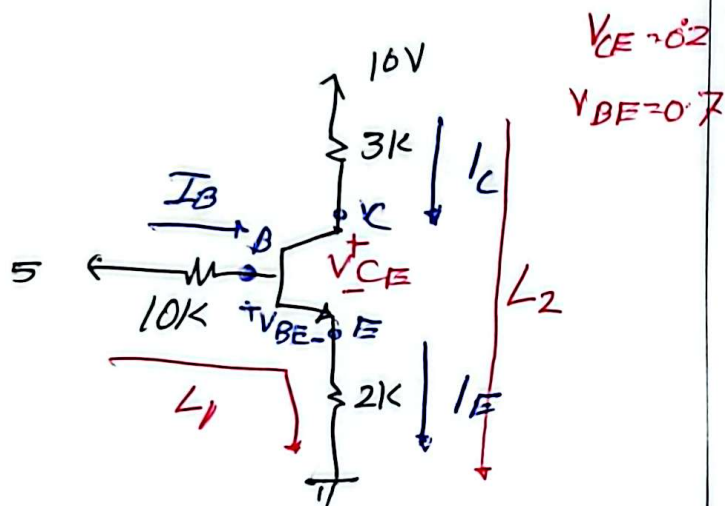
$$\therefore V_E = 4.096$$

$$V_{CE} = V_C - V_E$$

$$= 3.916 - 4.096$$

$$V_{CE} = -0.18 < 0.2$$

assume saturation.



KVL along L-1,

$$-10 + 3K I_C + V_{CE} + 2K I_E = 0 \quad \text{--- (I)}$$

KVL along L-2,

$$-5 + 10K I_B + V_{BE} + 2K I_E = 0 \quad \text{--- (II)}$$

&

$$I_E = I_C + I_B \quad \text{--- (III)}$$

$$\text{now, } I_E = \frac{V_E - 0}{2K}$$

$$\Rightarrow V_E = 2K I_E$$

$$\Rightarrow V_E = 2000 (\beta + 1) I_B$$

$$\therefore V_E = 2000 \times 101 \times 14.23 \times 10^{-6}$$

$$\therefore V_E = 2.8744V$$

$$V_{CE} = V_C - V_E$$

$$= 5.731 - 2.874$$

$$V_{CE} = 2.857 > 0.2$$

assumption correct.

note-

The circuits are almost same except the value of Base resistance.

That resulted in different operation.

now, we need V_C & V_E

$$\frac{5 - V_C}{10K} = I / I_E$$

$$I_E = \frac{V_E - 0}{2K}$$

$$V_C = 5 - 10K \times (\beta + 1) I_B$$

$$V_E = 2K \times (\beta + 1) I_B$$

$$\Rightarrow V_C = 5 - 10K \times 76 \times 4.61 \times 10^{-6}$$

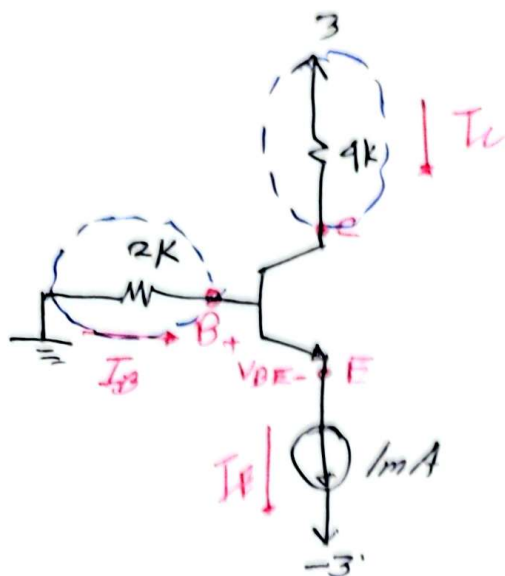
$$= 20K \times 76 \times 4.61 \times 10^{-6}$$

$$\therefore V_C = 1.4964$$

$$= 0.70072$$

$$\therefore V_{CE} = 0.795 > 0.2 \text{ Active mode}$$

6



$$\beta = 100$$

$$V_{BE} = 0.7 \text{ (Active)}$$

$$V_{BE} = 0.8 \text{ (sat)}$$

$$V_{CE} = 0.2 \text{ (sat)}$$

Note - since there's a current source at emitter terminal means current is flowing so. Not cut off.

Assume active.

$$I_E = 1\text{mA}$$

$$I_B = \frac{I_E}{\beta + 1}$$

$$= \frac{1\text{mA}}{100 + 1}$$

$$= 9.9\mu\text{A}$$

$$\frac{0 - V_B}{2\text{k}} = I_B$$

$$-V_B = 9.9\mu\text{A} \times 2\text{k}$$

$$\therefore V_B = 0.0198$$

$$\frac{3 - V_C}{4\text{k}} = I_C$$

$$\Rightarrow 3 - V_C = I_C \times 4\text{k}$$

$$\Rightarrow V_C = 3 - \beta I_B \times 4\text{k}$$

$$= 0.96$$

$$V_{BE} = 0.7$$

$$\Rightarrow V_B - V_E = 0.7$$

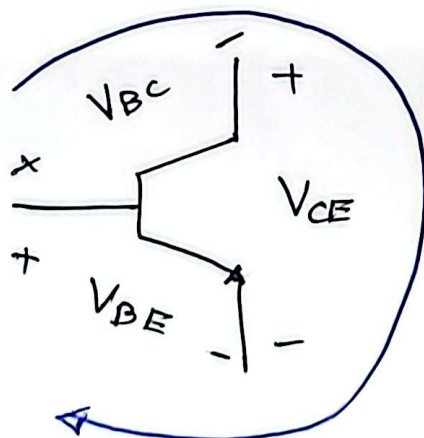
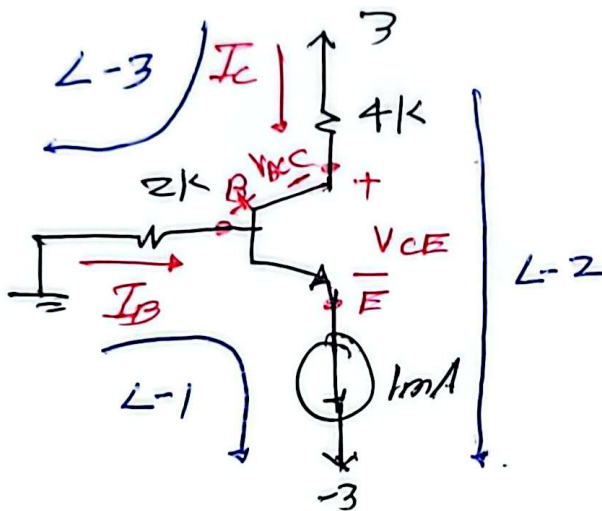
$$\Rightarrow V_E = V_B - 0.7$$

$$V_E = -0.6802$$

$$V_{CE} = V_C - V_E = -0.96 - (-0.6802)$$

$$V_{CE} = -0.2798 \not> 0.2$$

Assume saturation,



$$-V_{BE} + V_{BC} + V_{CE} = 0$$

$$\Rightarrow -0.8 + V_{BC} + 0.2 = 0$$

$$\therefore V_{BC} = 0.6$$

Now, along L-2,

$$-3 + 4k I_C - V_{BC} + 2k I_B = 0$$

$$\Rightarrow 4k I_C - 2k I_B = 3.6 \quad \text{--- (I)}$$

and we know,

$$I_C + I_B = 1 \quad \text{--- (II)}$$

$$I_C = 0.333$$

$$I_B = 0.667$$

$$\beta = \frac{I_C}{I_B} = 0.499 < \beta_{\text{active}}$$